

SWC-TCF — Smart Waste Collection Two-Commodity Flow exact model

Compact arc-based MILP: binary arcs x_{ij} , visit indicators g_i , fleet size k , and two continuous commodities per arc (f = waste on board, h = residual capacity) whose sum equals $Q \cdot x_{ij}$. Flow balance enforces capacity and eliminates subtours without exponential cut families. Solved by Gurobi / SCIP / HIGHS / CPLEX through OR-Tools, Pyomo or native gurobipy.

